# OPTIMAL SIZING OF MEAN-VARIANCE-LIQUIDATION LONG-SHORT PORTFOLIO ON DECENTRALIZED LENDING PROTOCOLS

We study the optimal initial sizing of a leveraged long–short position in a DeFi lending protocol under a survival-conditioned solvent-path payoff model: the position is terminated irrevocably when the collateral-to-liability health factor breaches one, and we model the payoff distribution on surviving paths together with the survival probability, without modelling the mechanics of liquidation itself. Under Kou double-exponential jump–diffusion dynamics, two classical tools — an exponential change of measure and a Laplace–resolvent reduction — combine to yield semi-analytic expressions for the conditional payoff moments and survival probability as explicit functions of the initial log health factor; these are new results for this problem and market. The expressions evaluate in milliseconds versus minutes for Monte Carlo, enabling smooth gradient-based optimisation. The same methodology extends directly to other barrier-terminated leveraged positions, including traditional margin accounts, FX carry trades with stop-loss triggers, and commodity spread positions. Using these expressions, we characterise how the optimal initial health buffer $h_0^\star$ trades off expected return against liquidation risk.

*Keywords*: DeFi lending; leveraged long–short; health factor; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent; optimal portfolio sizing

*JEL Classification*: G11; G12; C65

## 1. Introduction

DeFi lending protocols such as AAVE and Compound enable a leveraged long–short strategy: deposit a cryptocurrency as collateral, borrow a second asset, and sell it short. The position is governed by a real-time *health factor* — a borrowing-power-adjusted ratio of collateral to liability — and is liquidated automatically and irrevocably the instant that ratio falls below one. A practitioner must choose the initial coin holdings to balance two opposing forces: a more aggressive position amplifies expected returns on surviving paths but raises the probability of forced exit. Cryptocurrency markets compound the difficulty — daily log-returns of 10–20% are common and flash crashes can exceed 50% — making fat-tailed jump dynamics essential for realistic liquidation probabilities.

This paper provides the first exact, semi-analytic solution to the optimal sizing problem under a *survival-conditioned solvent-path payoff model*: the objective is defined on the payoff realised on paths that avoid liquidation, and the liquidation boundary is treated as an absorbing barrier[1].

We work under the Kou double-exponential jump–diffusion framework (Kou, 2002). The solution rests on two successive dimensional reductions. First, an exponential change of measure absorbs the short-leg price factor into the path-measure density, collapsing what is nominally a bivariate barrier problem to

[1] Internal mechanics — keeper fees, partial penalties, slippage — are outside the scope of the model.

a *univariate* first-exit problem for the log-ratio process under a tilted Lévy measure. Second, because exponential jump sizes are eigenfunctions of the Kou generator, the Laplace-domain resolvent of the first-exit problem reduces to a $3 \times 3$ linear system — the same size regardless of moment order — that is solved in closed form. Time-domain quantities follow by numerical Laplace inversion via the Talbot contour algorithm (Talbot, 1979). The entire optimisation therefore reduces to a scalar search over a single parameter, the *initial log health factor*, with the conditional expected payoff, its variance, and the liquidation probability all expressible as semi-analytic functions of that scalar. Unlike Monte Carlo, which suffers from sampling noise that becomes severe when few paths survive liquidation and requires fresh simulation at each candidate $h_0$, the semi-analytic expressions are noise-free and differentiable in $h_0$, reducing the optimization to a smooth scalar search; numerical experiments confirm a $\approx 55\times$ speed-up over Monte Carlo.

This paper sits at the intersection of the DeFi liquidation literature and the analytical first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations; these papers are empirical and do not address the ex-ante optimal sizing problem. Filling that gap requires tools from the first-passage literature for jump diffusions: Kou and Wang (2003) derive explicit Laplace-transform results for first-passage times under double-exponential jump diffusion, and Sepp (2004) extends this to single- and double-barrier option pricing, reducing the resolvent to a finite algebraic system — the same structural advantage exploited here. The difference is that the payoff is a barrier-killed long–short ratio, not a single-asset barrier option; the change-of-measure reduction is the key step that bridges the two settings. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs, while Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our setting is static and the binding constraint is an endogenous first-passage event rather than a variance bound.

The paper makes the following contributions. First, we show that the portfolio optimisation reduces to a one-dimensional scalar problem parameterised by the initial log health factor, with all performance quantities expressible as explicit functions of that single parameter (Section 2). Second, we derive a semi-analytical Laplace–resolvent for killed moments of all admissible orders under the exponentially tilted measure (Section 5): because the payoff kink lies strictly below the liquidation barrier whenever the borrowing power is strictly less than one, the source term is a polynomial on the entire survival domain and the resolvent reduces to a three-by-three linear system with no interface conditions, regardless of moment order. Third, we derive the survival probability by the same resolvent method at zero tilt (Section 5, Subsection 5.2).

The remainder of the paper is organised as follows. Section 2 introduces the position, health factor, payoff, and the one-dimensional reduction to a scalar optimisation. Section 3 specifies the asset dynamics under the Kou model; Section 3.2 introduces the tilted-measure machinery. Section 4 shows how the change of measure decouples the short-leg factor from the barrier problem, reducing each moment to a one-dimensional killed expectation, and states the objective in terms of those moments. Section 5 derives the respective resolvents. Section 6 presents numerical results, comparing the Laplace–resolvent method against Monte Carlo simulation across a range of initial log health factor values and verifying the analytical no-shorting limit.

## 2. Problem Setup

We consider a leveraged long–short portfolio in which an investor holds a long position in asset $S_1$ and a short position in asset $S_2$, financed so that the net initial value of the book equals one unit of account. The strategy is subject to a risk-management rule that forces liquidation if the long leg falls sufficiently relative to the short leg, reflecting margin requirements or stop-loss constraints common in practice. The objective is to choose the initial coin holdings so as to optimise a performance criterion defined on the terminal payoff, conditional on surviving the liquidation threshold.

### 2.1. Terminal payoff of the long-short portfolio

Let $(\Omega, \mathscr{F}, (\mathscr{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices $S_1, S_2$ evolve as

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \qquad X_{i,0} = 0, \quad i = 1, 2,$$

where $X_1, X_2$ are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset $S_1$ (long leg) and $n_2 > 0$ units of asset $S_2$ (short leg).

Let

$$V_{1,t} := n_1 S_{1,t}, \qquad V_{2,t} := n_2 S_{2,t}$$

denote the *collateral value* (long leg) and the *borrow value* (outstanding liability) at time $t$. Fix $b \in (0,1)$, the *collateral factor* of asset $S_1$ (called *liquidation threshold* in AAVE v3 terminology): the protocol requires the $b$-adjusted collateral to cover the borrow value at all times,

$$bV_{1,t} \geq V_{2,t}.$$

The *health factor* summarises this margin condition as a single ratio,

$$H_t := \frac{bV_{1,t}}{V_{2,t}} = \frac{bn_1 S_{1,t}}{n_2 S_{2,t}}, \qquad h_t := \log H_t, \tag{1}$$

where $h_t$ is the *log health factor*. The solvency condition is $H_t \geq 1$ ($h_t \geq 0$); breaching it is equivalent to the borrow value exceeding the $b$-adjusted collateral. Liquidation is triggered at the first such instant:

$$\tau := \inf\{t \geq 0 : \ H_t < 1\} = \inf\{t \geq 0 : \ h_t < 0\}.$$

A smaller $b$ allows a larger borrow-to-collateral ratio before the boundary is breached, giving the position more room to absorb adverse price moves; a larger $b$ triggers earlier forced exit and limits the protocol's risk of bad debt.

Since

$$h_t = \log\left(\frac{bn_1 S_{1,0}}{n_2 S_{2,0}}\right) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\left(\frac{bn_1 S_{1,0}}{n_2 S_{2,0}}\right), \qquad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : \ Z_t < -h_0\}.$$

The one-dimensional process $Z$ thus inherits all the relevant risk: the position survives if and only if $Z$ stays above the absorbing level $-h_0$. The parameter $h_0 = \log H_0$ is therefore the *initial log-cushion* — the log-distance of the initial health factor from the liquidation boundary $H = 1$. A larger $h_0$ means the position starts further from forced exit; $h_0 \to 0^+$ means the position starts almost at the boundary with negligible cushion.

The gross profit at horizon $T$, contingent on survival, is

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T})^+ \mathbf{1}_{\{\tau > T\}}.$$

The positive part reflects that the investor closes the position only if it is in profit at $T$; the indicator $\mathbf{1}_{\{\tau>T\}}$ reflects that the position must have survived to maturity. This paper is a *survival-conditioned solvent-path payoff model*: it characterises the distribution of $\Pi_T$ on the event $\{\tau > T\}$ and the probability of that event, with the boundary $H_t = 1$ treated as an absorbing state. The liquidation mechanics are intentionally outside the model scope; the remark below clarifies what is omitted and why the absorbing-barrier proxy is nonetheless appropriate for position-sizing purposes.

*Remark* 1 (Liquidation mechanics). When $H_t$ hits one, a DeFi lending protocol does not simply extinguish the position: it initiates a protocol-specific liquidation procedure executed by third-party keepers. In AAVE v3, a keeper may close up to 50% of the debt (100% if the health factor falls below a secondary threshold), claiming a *liquidation bonus* of 5–15% of the seized collateral as compensation; Compound uses a fixed close factor with a similar bonus structure (Qin et al., 2021; Heimbach and Huang, 2024).

The execution price received by the keeper — and hence the residual recovery value for the borrower — is itself uncertain. Liquidations in DeFi are settled on-chain via decentralised exchanges: in stressed markets, the available liquidity may be insufficient to absorb a large collateral sale without significant slippage, compressing the borrower's recovery further (Qin et al., 2021). The mechanics therefore depend on the specific protocol version, the pool liquidity at the moment of liquidation, the gas price environment, and the speed of keeper response — none of which are captured by a single-barrier first-passage model.

For these reasons we assign zero payoff to liquidated paths: $\Pi_T := 0$ on $\{\tau \le T\}$. This is a *conservative proxy* — it understates the expected residual value whenever the borrower retains a fractional recovery, and overstates the loss relative to protocols that execute partial liquidations. Its practical justification is two-fold. First, the borrower typically has no control over the execution details once the boundary is breached; from an ex-ante sizing perspective the liquidation payoff is a zero-mean random variable with uncertain sign, and setting it to zero is the risk-neutral midpoint. Second, the penalty term $\delta\, p_{\mathrm{liq}}$ in the objective of (5) provides a separate lever to price liquidation risk at whatever level the investor deems appropriate, without requiring an explicit model of the post-liquidation cash flow. The fragility of DeFi lending protocols under large price shocks (Chiu et al., 2022) further supports a conservative treatment of the post-liquidation value.

Using the definition of $H_T$, one obtains the following factorization.

**Lemma 2.1** (Payoff factorization)**.**

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} \left(e^{h_0+Z_T} - b\right)^+ \mathbf{1}_{\{\tau>T\}}. \tag{2}$$

This separates the payoff into a numeraire factor driven by $S_2$, and a barrier-killed option on the log-ratio $Z$ with moneyness controlled by $h_0$.

### *2.2. Optimization setup*

The two coin holdings $(n_1, n_2)$ represent two degrees of freedom. A normalisation reduces this to a single scalar, and the structure of the payoff then identifies $h_0$ as the natural optimisation variable.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1 + x)/x$. Solving for $x$,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \qquad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \tag{3}$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

The initial gross leverage — long-leg value per unit of equity — follows directly from (3):

$$L_0 := n_1(h_0)\, S_{1,0} = \frac{e^{h_0}}{e^{h_0} - b}. \tag{4}$$

$L_0$ is strictly decreasing in $h_0$: as $h_0 \downarrow \ln b$, leverage $L_0 \uparrow \infty$; as $h_0 \to \infty$, leverage $L_0 \downarrow 1$ (the short leg vanishes and the position becomes unlevered). Equivalently, the borrow-to-equity ratio is $L_0 - 1 = b/(e^{h_0} - b)$, which also collapses the initial health factor back to $H_0 = e^{h_0}$ via $H_0 = b L_0/(L_0 - 1)$. Choosing $h_0$ is therefore equivalent to choosing the initial leverage.

Since every quantity of interest — the expected payoff, its variance, and the liquidation probability — is determined by $h_0$ through (3) and the killed process $(Z, \tau)$, the portfolio problem reduces to a one-dimensional optimisation:

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0),$$

where $\Phi$ is a scalar performance criterion, for example the Sharpe ratio, expected return subject to a liquidation probability constraint, or a mean–variance objective. The optimal holdings are then recovered by substituting $h_0^\star$ back into (3).

The specific criterion studied in this paper is a penalised conditional mean–variance objective. Let $\mathbb{E}[\cdot \mid \tau > T]$ denote expectation conditional on non-liquidation. The objective is

$$\Phi(h_0) := \mathbb{E}[\Pi_T \mid \tau > T] - \tfrac{\alpha}{2} \operatorname{Var}(\Pi_T \mid \tau > T) - \delta\, p_{\text{liq}}(T, 0; h_0), \tag{5}$$

where $\alpha \geq 0$ is a risk-aversion coefficient and $\delta \geq 0$ penalises the probability of forced liquidation. The first two terms reward a high risk-adjusted return on the surviving paths; the third term discourages positions so aggressive that liquidation becomes likely. The interplay between the three terms means that the optimal $h_0^\star$ — which we call the *optimal initial health buffer* — balances leverage against downside protection: a lower $h_0$ implies higher leverage, a larger conditional upside on surviving paths, but a higher probability of forced liquidation; a higher $h_0$ provides more cushion above the barrier, reducing liquidation risk at the cost of a smaller conditional payoff. We do not refer to $h_0^\star$ as "optimal leverage" because leverage is a derived quantity (4) that is not directly controlled; the investor chooses the health buffer and leverage follows.

The remainder of the paper is devoted to computing the building blocks of $\Phi$: the killed moments $m_k(T; h_0)$ and the survival probability $p_{\text{surv}}(T, 0; h_0)$, each as a closed-form function of $h_0$, evaluated semi-analytically via the Laplace–resolvent method developed in Section 5.

## 3. Asset Dynamics

### *3.1. Bivariate Kou model*

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$\mathrm{d}X_{i,t} = \mu_i^{\text{eff}}\, \mathrm{d}t + \sigma_i\, \mathrm{d}B_{i,t} + \mathrm{d}\left( \sum_{n=1}^{N_{i,t}} J_i^{(n)} \right), \qquad i = 1, 2, \tag{6}$$

where $B_1, B_2$ are correlated standard Brownian motions with $\mathrm{d}\langle B_1, B_2\rangle_t = \rho\,\mathrm{d}t$, $N_i$ is a Poisson process with intensity $\lambda_i$ independent of $(B_1, B_2)$, and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \qquad \varphi_{J_i}(u) = \frac{p_i}{1 - iu\eta_{i,+}} + \frac{1-p_i}{1 + iu\eta_{i,-}}, \tag{7}$$

$$M_i(s) = \frac{p_i}{1 - s\eta_{i,+}} + \frac{1-p_i}{1 + s\eta_{i,-}}, \qquad s \in \left(-\tfrac{1}{\eta_{i,-}}, \tfrac{1}{\eta_{i,+}}\right). \tag{8}$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator $\chi_i$ defined below to be finite.

The parameter $\mu_i$ is the *continuously compounded expected asset-price growth rate*, i.e. $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1 - \eta_{i,+}} + \frac{1-p_i}{1 + \eta_{i,-}} - 1. \tag{9}$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^{\mathrm{eff}} + \frac{1}{2}\sigma_i^2 + \lambda_i \chi_i])$, matching to $e^{\mu_i t}$ gives the log-price drift coefficient

$$\mu_i^{\mathrm{eff}} := \mu_i - \tfrac{1}{2}\sigma_i^2 - \lambda_i \chi_i. \tag{10}$$

All analytical and numerical objects in the sequel use $\mu_i^{\mathrm{eff}}$.

The bivariate process $(X_1, X_2)$ is a Lévy process with joint characteristic exponent

$$\begin{aligned} \Psi(u,v) := {} & i(\mu_1^{\mathrm{eff}} u + \mu_2^{\mathrm{eff}} v) - \tfrac{1}{2}\Big(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv\Big) \\ & + \lambda_1\big(\varphi_{J_1}(u) - 1\big) + \lambda_2\big(\varphi_{J_2}(v) - 1\big), \end{aligned} \tag{11}$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t} + vX_{2,t})}] = e^{t\Psi(u,v)}$.

Table 1 collects all model parameters and their roles.

### *3.2. Tilted Dynamics*

For each integer $k \geq 0$, define the $k$-tilted measure $\mathbb{P}^{(k)}$ by

$$\left.\frac{\mathrm{d}\mathbb{P}^{(k)}}{\mathrm{d}\mathbb{P}}\right|_{\mathscr{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp\left(kX_{2,t} - t\,\Psi(0, -ik)\right), \qquad k\eta_{2,+} < 1.$$

**Lemma 3.1** (Lévy exponent under $\mathbb{P}^{(k)}$)**.** *Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)} s + \tfrac{1}{2}\sigma_Z^2 s^2 + \lambda_1\big(M_1(s) - 1\big) + \lambda_2\big(M_2(k-s) - M_2(k)\big), \tag{12}$$

*where*

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \qquad \mu_Z^{(k)} := (\mu_1^{\mathrm{eff}} - \mu_2^{\mathrm{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

*In particular, the generator satisfies*

$$\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s)\, e^{sz}. \tag{13}$$

| Symbol | Asset | Economic meaning / role | Constraint |
|---|---|---|---|
| $\mu_i$ | $i=1,2$ | Continuously compounded expected price growth rate: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$ | |
| $\sigma_i$ | $i=1,2$ | Diffusion volatility of log-price $X_i$ | $\sigma_i > 0$ |
| $\rho$ | — | Correlation between the two Brownian drivers | $\lvert\rho\rvert < 1$ |
| $\lambda_i$ | $i=1,2$ | Jump arrival intensity (expected jumps per unit time) | $\lambda_i > 0$ |
| $p_i$ | $i=1,2$ | Probability that a jump in $X_i$ is upward | $p_i \in (0,1)$ |
| $\eta_{i,+}$ | $i=1,2$ | Mean size of upward jumps in $X_i$ | $0 < \eta_{i,+} < 1$ |
| $\eta_{i,-}$ | $i=1,2$ | Mean size of downward jumps in $X_i$ | $\eta_{i,-} > 0$ |
| $\chi_i$ | $i=1,2$ | Price-jump compensator $M_i(1)-1 = \mathbb{E}[e^{J_i}-1]$; enters $\mu_i^{\text{eff}}$ via (10) | |
| $\mu_i^{\text{eff}}$ | $i=1,2$ | Actual drift coefficient in the log-price SDE (6) | |
| $\Psi(u,v)$ | — | Joint characteristic exponent; $e^{T\Psi(0,-ik)} = \mathbb{E}[e^{kX_{2,T}}]$ appears in the moment formula | |
| $S_{i,0}$ | $i=1,2$ | Initial price of asset $i$ | $S_{i,0} > 0$ |
| $b$ | — | Borrowing power of the collateral (AAVE liquidation threshold) | $b \in (0,1)$ |
| $T$ | — | Investment horizon | $T > 0$ |
| $n_i$ | $i=1,2$ | Number of coins held: $n_1 > 0$ long in $S_1$, $n_2 > 0$ short in $S_2$ | $n_i > 0$ |
| $H_t$ | — | Health factor: $bn_1S_{1,t}/(n_2S_{2,t})$; liquidated when $H_t < 1$ | |
| $h_t$ | — | Log health factor: $h_t = \log H_t = h_0 + Z_t$ | |
| $h_0$ | — | Initial log health factor; the optimisation variable; determines $n_i$ via (3) | $h_0 > \log b$ |
| $\tau$ | — | Liquidation time: $\tau = \inf\{t \ge 0 : H_t < 1\}$ | |
| $\Pi_T$ | — | Terminal payoff: $(n_1S_{1,T} - n_2S_{2,T})^+ \mathbf{1}_{\{\tau > T\}}$ | |

Table 1: Model parameters and derived quantities. The first block contains freely chosen Kou model inputs; the second block contains derived model quantities; the third block contains portfolio and contract inputs; the fourth block contains derived portfolio quantities.

*Proof. Step 1: Laplace transform under $\mathbb{P}^{(k)}$.* Because $Z_t = X_{1,t} - X_{2,t}$ and the density on $\mathscr{F}_t$ equals $e^{kX_{2,t} - t\Psi(0,-ik)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}\left[e^{sZ_t}\right] = e^{-t\Psi(0,-ik)}\,\mathbb{E}_{\mathbb{P}}\left[e^{sX_{1,t}+(k-s)X_{2,t}}\right] = e^{t[\Psi(-is,-i(k-s))-\Psi(0,-ik)]}, \tag{14}$$

where we used $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t}+\beta X_{2,t}}] = e^{t\Psi(-i\alpha,-i\beta)}$. Since the Radon–Nikodým density is exponential-linear in $(X_{1,t}, X_{2,t})$, Girsanov's theorem for Lévy processes guarantees that $Z$ remains a Lévy process under $\mathbb{P}^{(k)}$ with exponent

$$\psi_Z^{(k)}(s) = \Psi(-is,\,-i(k-s)) - \Psi(0,-ik). \tag{15}$$

*Step 2: Expand and simplify.* Substituting (11) with $(u,v) = (-is, -i(k-s))$, and noting $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$, and $\varphi_{J_i}(-i\cdot) = M_i(\cdot)$:

$$\begin{aligned}\Psi(-is,-i(k-s)) = \mu_1^{\text{eff}}s + \mu_2^{\text{eff}}(k-s) + \tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2(k-s)^2 + \rho\sigma_1\sigma_2 s(k-s)\\ + \lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s)-1).\end{aligned} \tag{16}$$

Setting $s=0$ in (16) gives $\Psi(0,-ik) = \mu_2^{\text{eff}}k + \frac{1}{2}\sigma_2^2k^2 + \lambda_2(M_2(k)-1)$. Subtracting and collecting terms by type:

*Drift.* $\mu_1^{\text{eff}}s + \mu_2^{\text{eff}}(k-s) - \mu_2^{\text{eff}}k = (\mu_1^{\text{eff}} - \mu_2^{\text{eff}})s.$

*Diffusion.* Expanding $(k-s)^2 = k^2 - 2ks + s^2$ and $(k-s) = k - s$:

$$\begin{aligned}&\tfrac{1}{2}\sigma_1^2 s^2 + \tfrac{1}{2}\sigma_2^2(k-s)^2 + \rho\sigma_1\sigma_2 s(k-s) - \tfrac{1}{2}\sigma_2^2 k^2 \\ &\quad = \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s = \tfrac{1}{2}\sigma_Z^2 s^2 - k(\sigma_2^2 - \rho\sigma_1\sigma_2)s.\end{aligned}$$

*Jumps.* $\lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s)-1) - \lambda_2(M_2(k)-1) = \lambda_1(M_1(s)-1) + \lambda_2(M_2(k-s) - M_2(k))$.

Combining, the drift coefficient is $\mu_Z^{(k)} := (\mu_1^{\text{eff}} - \mu_2^{\text{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$, which establishes (12). The eigenrelation (13) follows because for any Lévy process with exponent $\psi$, the generator satisfies $\mathscr{L}e^{sz} = \psi(s)\,e^{sz}$, as can be seen by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\,\psi(s)}$ with respect to $t$ at $t = 0$. □

This eigenrelation is the key identity used in Section 5 to build the resolvent equation.

*Remark* 2 (Superscript $(k)$ convention). A superscript $(k)$ indicates evaluation under $\mathbb{P}^{(k)}$. The untilted case $k = 0$ recovers the original dynamics of $Z$ under $\mathbb{P}$. For $k \geq 1$, the $k$-th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$; the constraint $k\eta_{2,+} < 1$ ensures the tilted measure and the characteristic root structure remain well-defined.

For the Kou structure, the tilt transforms the jump rates of $Z$ into the $k$-dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of $X_2$ appear as downward jumps of $Z$: the tilt by $e^{kX_2}$ changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$.

**Definition 3.1** (Phase operators)**.** For $j = 1, 2$, the *phase operators* associated with the tilted jump kernels of $Z$ under $\mathbb{P}^{(k)}$ are

$$K_{j,+}^{(k)}[f](z) := \int_0^\infty f(z+y)\,r_{j,+}^{(k)}e^{-r_{j,+}^{(k)}y}\,\mathrm{d}y, \qquad K_{j,-}^{(k)}[f](z) := \int_{-\infty}^0 f(z+y)\,r_{j,-}^{(k)}e^{r_{j,-}^{(k)}y}\,\mathrm{d}y.$$

This construction originates in Kou and Wang (2003) for first-passage problems under the double-exponential model, and is used by Sepp (2004) to reduce double-barrier pricing to a finite linear system. For any $\gamma$ with $\gamma < r_{j,+}^{(k)}$ and $\gamma > -r_{j,-}^{(k)}$, evaluating the integrals directly gives the eigenfunction identity

$$K_{j,+}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma}\,e^{\gamma z}, \qquad K_{j,-}^{(k)}[e^{\gamma\cdot}](z) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma}\,e^{\gamma z}. \tag{17}$$

Each phase operator maps an exponential to a scalar multiple of the same exponential, which is the key algebraic identity exploited in Section 5 to convert the resolvent PIDE into a finite-dimensional linear system.

Table 2 collects the quantities introduced in this subsection.

## 4. Moments via Measure Change

Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is intractable: the payoff couples the joint process $(X_1, X_2)$ to the barrier event $\{\tau > T\}$ in a way that is neither separable nor analytically tractable. An exponential change of measure resolves this by tilting the path measure by $e^{kX_{2,T}}$, which absorbs the short-leg factor into the density and leaves a purely one-dimensional killed expectation for the log-ratio $Z = X_1 - X_2$ under the $k$-tilted measure $\mathbb{P}^{(k)}$. Since $Z$ remains a Lévy process under $\mathbb{P}^{(k)}$, no information about the joint

| Symbol | Definition | Role |
|---|---|---|
| $\mathbb{P}^{(k)}$ | $\mathrm{d}\mathbb{P}^{(k)}/\mathrm{d}\mathbb{P} \propto e^{kX_{2,t}}$ | Measure under which moment reduction is carried out |
| $k$ | Non-negative integer, $k\eta_{2,+} < 1$ | Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability |
| $\psi_Z^{(k)}(s)$ | Lévy exponent of $Z$ under $\mathbb{P}^{(k)}$ | Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent |
| $\sigma_Z^2$ | $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ | Variance of the Brownian part of $Z$; appears in $\psi_Z^{(k)}$ |
| $\mu_Z^{(k)}$ | $(\mu_1^{\text{eff}} - \mu_2^{\text{eff}}) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$ | Drift of $Z$ under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$ |
| $\mathscr{L}^{(k)}$ | Generator of $Z$ under $\mathbb{P}^{(k)}$ | $\mathscr{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}$; used in the resolvent PIDE |
| $r_{j,\pm}^{(k)}$ | Phase rates (see above) | Exponential rates of the jump components of $Z$ under $\mathbb{P}^{(k)}$ |
| $K_{j,\pm}^{(k)}$ | Phase operators (see above) | Convert jump integrals to scalars on exponential ansatz |

Table 2: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from $X_1$ jumps; $j = 2$ to that from $X_2$ jumps (sign-reversed because $Z = X_1 - X_2$).

dynamics is lost: the tilted characteristic exponent encodes the full marginal law of $Z$. The $k$-th moment then factorises as the known scalar $e^{T\Psi(0,-ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ times the barrier-killed expectation $m_k(T;h_0)$ under $\mathbb{P}^{(k)}$, which is computed via the Laplace–resolvent of Section 5.

Define

$$g_k(z;h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, and $b \in (0,1)$ gives $z_*(h_0) := \ln b - h_0 < -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$. The positive part in (2) is therefore redundant on the survival set, and $g_k$ is a polynomial in $e^{Z_T}$. From (2),

$$\Pi_T^k = \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}}\, g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}.$$

$$\mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \left(\frac{n_2}{b}\right)^k S_{2,0}^k\, e^{T\Psi(0,-ik)}\, \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}\Big]. \tag{18}$$

**Proposition 4.1** (Moments of any order)**.** *Let $k \geq 1$ be an integer satisfying $k\eta_{2,+} < 1$. Define*

$$m_k(T;h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}\Big[g_k(Z_T;h_0)\, \mathbf{1}_{\{\tau>T\}}\Big].$$

*Then*

$$M_k(h_0) := \mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T;h_0),$$

*where $m_k(T;h_0) = U^{(k)}(T,0;h_0)$ is computed via the Laplace–resolvent method of Section 5. The maximum admissible order is $k_{\max} = \lfloor 1/\eta_{2,+} \rfloor - 1$.*

Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since $\Pi_T$ already contains the survival indicator $\mathbf{1}_{\{\tau>T\}}$, the conditional $k$-th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{M_k(h_0)}{p_{\text{surv}}(h_0)}, \qquad k \geq 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)}, \qquad \mathrm{Var}(\Pi_T \mid \tau > T) = \frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2.$$

The mean–variance–liquidation objective is therefore

$$\Phi(h_0) = \frac{M_1(h_0)}{p_{\text{surv}}(h_0)} - \frac{\alpha}{2}\left[\frac{M_2(h_0)}{p_{\text{surv}}(h_0)} - \left(\frac{M_1(h_0)}{p_{\text{surv}}(h_0)}\right)^2\right] - \delta\, p_{\text{liq}}(h_0), \tag{19}$$

and the portfolio optimization reduces to the one-dimensional problem

$$\max_{h_0\in(\ln b,\infty)} \Phi(h_0),$$

after which the optimal coin holdings are recovered from (3).

*Remark* 3 (No-shorting limit). As $h_0 \to \infty$ the holding map gives $n_2S_{2,0} \to 0$ and $n_1S_{1,0} \to 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every $k \geq 1$,

$$\lim_{h_0\to\infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which provides a useful diagnostic: the formula for $M_k(h_0)$ must satisfy this limit exactly. The $k = 1$ case gives $e^{\mu_1 T}$ by the drift normalisation (10).

The remainder of the analysis is devoted to computing $m_k(T;h_0)$ for all admissible $k \geq 1$ and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

## 5. Resolvent Equations for Moments and Survival Probability

With the moment problem reduced to a one-dimensional killed expectation under $\mathbb{P}^{(k)}$, the function $U(t,z) = \mathbb{E}_z^{(k)}[g_k(Z_t;h_0)\mathbf{1}_{\{\tau>t\}}]$ satisfies a PIDE in $(t,z)$. Taking the Laplace transform in $t$ converts this to a stationary integro-differential equation in $z$ alone, with the Laplace parameter acting as a discount rate. The Kou structure then reduces it further: because exponentials are eigenfunctions of the generator, every integro-differential term acts algebraically on an exponential ansatz, and the resolvent collapses to a finite-dimensional linear system. Three boundary conditions at the absorbing barrier close the $3\times 3$ system. The survival probability follows by the same architecture at zero tilt ($k = 0$). This section derives both resolvents in closed form.

### *5.1. Killed Moment Resolvent*

Fix an integer $k \geq 1$ with $k\eta_{2,+} < 1$. Define the killed moment function

$$U(t,z) := \mathbb{E}_z^{(k)}\left[g_k(Z_t;h_0)\,\mathbf{1}_{\{\tau(h_0)>t\}}\right], \qquad z > -h_0,$$

with $U(t,z) = 0$ for $z \leq -h_0$. This satisfies the PIDE

$$\partial_t U = \mathscr{L}^{(k)}U, \qquad z > -h_0,$$

with initial condition $U(0,z) = g_k(z;h_0)$. Define the Laplace transform

$$\widehat{U}(q,z) := \int_0^\infty e^{-qt}U(t,z)\,\mathrm{d}t, \qquad q > 0.$$

Applying it to both sides of the PIDE proceeds in two steps.

*Left-hand side.* Integrate by parts in $t$:

$$\int_0^\infty e^{-qt}\,\partial_t U(t,z)\,\mathrm{d}t = \Big[e^{-qt}U(t,z)\Big]_0^\infty + q\int_0^\infty e^{-qt}U(t,z)\,\mathrm{d}t = -U(0,z) + q\widehat{U}(q,z),$$

where the boundary term at $t=\infty$ vanishes by integrability.

*Right-hand side.* Since $\mathscr{L}^{(k)}$ acts only in $z$, it commutes with the $t$-integral:

$$\int_0^\infty e^{-qt}\,\mathscr{L}^{(k)}U(t,z)\,\mathrm{d}t = \mathscr{L}^{(k)}\widehat{U}(q,z).$$

Equating and substituting the initial condition $U(0,z) = g_k(z;h_0)$ gives the resolvent equation

$$(q-\mathscr{L}^{(k)})\widehat{U}(q,z) = g_k(z;h_0), \qquad z > -h_0, \tag{20}$$

with $\widehat{U}(q,z) = 0$ for $z \le -h_0$.

**Assumption 5.1** (Generic six-root regime)**.** Fix an integer $k \ge 0$ satisfying $k\eta_{2,+} < 1$, and let $q > 0$. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1(q),\dots,\gamma_6(q)$, ordered so that

$$\mathrm{Re}(\gamma_{1,2,3}) > 0, \qquad \mathrm{Re}(\gamma_{4,5,6}) < 0;$$

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0,\dots,k$.

Since $b \in (0,1)$ and $z > -h_0$ together imply $e^{h_0+z} > 1 > b$, the payoff kernel expands as a polynomial in $e^z$:

$$g_k(z;h_0) = \sum_{j=0}^{k} c_{k,j}(h_0)\,e^{jz}, \qquad c_{k,j}(h_0) := \binom{k}{j}(-1)^{k-j}b^{k-j}e^{jh_0}. \tag{21}$$

**Lemma 5.1** (Particular solution)**.** *Under Assumption 5.1(iii),*

$$\widehat{U}_{\mathrm{part}}(q,z;h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}\,e^{jz} \tag{22}$$

*is a particular solution to* (20)*.*

*Proof.* Substitute the forcing expansion into the right-hand side of (20):

$$(q-\mathscr{L}^{(k)})\widehat{U}_{\mathrm{part}} = \sum_{j=0}^{k} c_{k,j}(h_0)\,e^{jz}.$$

Since the sum is finite and $\mathscr{L}^{(k)}$ is linear, it suffices to find, for each $j$, a function $v_j$ satisfying

$$(q-\mathscr{L}^{(k)})v_j(z) = c_{k,j}(h_0)\,e^{jz}.$$

Propose the exponential ansatz $v_j(z) = A_j\,e^{jz}$. By the eigenrelation (13),

$$\mathscr{L}^{(k)}[A_j e^{jz}] = A_j\,\psi_Z^{(k)}(j)\,e^{jz},$$

so

$$(q-\mathscr{L}^{(k)})[A_j e^{jz}] = A_j\big(q-\psi_Z^{(k)}(j)\big)e^{jz}.$$

Matching to $c_{k,j}(h_0)\,e^{jz}$ and using Assumption 5.1(iii) to ensure $q-\psi_Z^{(k)}(j)\neq 0$ gives

$$A_j = \frac{c_{k,j}(h_0)}{q-\psi_Z^{(k)}(j)}.$$

Summing over $j=0,\ldots,k$ yields (22). $\square$

**Lemma 5.2** (Homogeneous solution)**.** *Under Assumption 5.1(i)–(ii), the general solution to $(q-\mathscr{L}^{(k)})u = 0$ on $z>-h_0$ that is admissible at $+\infty$ is*

$$\widehat{U}_{\text{hom}}(q,z;h_0) = \sum_{m=1}^{3} B_m^{(k)}(q;h_0)\, e^{\gamma_{m+3}(q)(z+h_0)}, \tag{23}$$

*where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants.*

*Proof. Step 1: exponential ansatz.* Seek solutions of the form $u(z)=e^{\gamma z}$. By the eigenrelation (13),

$$\mathscr{L}^{(k)} e^{\gamma z} = \psi_Z^{(k)}(\gamma)\, e^{\gamma z},$$

so $(q-\mathscr{L}^{(k)})e^{\gamma z} = (q-\psi_Z^{(k)}(\gamma))e^{\gamma z}$. This vanishes if and only if $\gamma$ satisfies the characteristic equation

$$\psi_Z^{(k)}(\gamma) = q.$$

*Step 2: root count.* The exponent $\psi_Z^{(k)}$ is the sum of a quadratic term $\frac{1}{2}\sigma_Z^2\gamma^2$ (non-degenerate by Assumption 5.1(i)) and rational terms from the Kou jumps. Clearing denominators yields a degree-six polynomial in $\gamma$, so there are exactly six roots $\gamma_1,\ldots,\gamma_6$ (counted with multiplicity). Under Assumption 5.1(ii) all six are simple and split as $\mathrm{Re}(\gamma_{1,2,3})>0$ and $\mathrm{Re}(\gamma_{4,5,6})<0$.

*Step 3: admissibility.* On the half-line $z>-h_0$, the solution must remain bounded as $z\to+\infty$. Since $e^{\gamma z}\to\infty$ whenever $\mathrm{Re}(\gamma)>0$, the roots $\gamma_1,\gamma_2,\gamma_3$ are inadmissible. Only $\gamma_4,\gamma_5,\gamma_6$ are retained.

*Step 4: centering and general solution.* Writing $e^{\gamma_{m+3}z} = e^{\gamma_{m+3}(z+h_0)}\cdot e^{-\gamma_{m+3}h_0}$ and absorbing the constant factor into the free coefficient, the general admissible homogeneous solution is

$$\widehat{U}_{\text{hom}}(q,z;h_0) = \sum_{m=1}^{3} B_m^{(k)}(q;h_0)\, e^{\gamma_{m+3}(q)(z+h_0)},$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants to be determined by the boundary conditions. $\square$

The general solution on $z>-h_0$ is then

$$\widehat{U}(q,z;h_0) = \widehat{U}_{\text{part}}(q,z;h_0) + \widehat{U}_{\text{hom}}(q,z;h_0).$$

**Proposition 5.3** (Barrier resolvent)**.** *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \tag{24}$$

*the boundary conditions at $z=-h_0^+$ uniquely determine $\mathbf{B}^{(k)}(q;h_0) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ via the linear system*

$$\mathbf{M}_{\text{bar}}^{(k)}(q)\,\mathbf{B}^{(k)}(q;h_0) = -\mathbf{b}^{(k)}(q;h_0),$$

*where the $3\times 3$ barrier matrix is*

$$\mathbf{M}^{(k)}_{\mathrm{bar}}(q)=\begin{pmatrix}1 & 1 & 1\\ \dfrac{r^{(k)}_{1,-}}{r^{(k)}_{1,-}+\gamma_4(q)} & \dfrac{r^{(k)}_{1,-}}{r^{(k)}_{1,-}+\gamma_5(q)} & \dfrac{r^{(k)}_{1,-}}{r^{(k)}_{1,-}+\gamma_6(q)}\\ \dfrac{r^{(k)}_{2,-}}{r^{(k)}_{2,-}+\gamma_4(q)} & \dfrac{r^{(k)}_{2,-}}{r^{(k)}_{2,-}+\gamma_5(q)} & \dfrac{r^{(k)}_{2,-}}{r^{(k)}_{2,-}+\gamma_6(q)}\end{pmatrix}$$

*and the forcing vector is*

$$\mathbf{b}^{(k)}(q;h_0)=\begin{pmatrix}\widehat{U}_{\mathrm{part}}(q,-h_0;h_0)\\ K^{(k)}_{1,-}\left[\widehat{U}_{\mathrm{part}}(q,\cdot;h_0)\right](-h_0)\\ K^{(k)}_{2,-}\left[\widehat{U}_{\mathrm{part}}(q,\cdot;h_0)\right](-h_0)\end{pmatrix}=\begin{pmatrix}\displaystyle\sum_{j=0}^{k}\frac{c_{k,j}(h_0)}{q-\psi^{(k)}_Z(j)}\,e^{-jh_0}\\ \displaystyle\sum_{j=0}^{k}\frac{c_{k,j}(h_0)}{q-\psi^{(k)}_Z(j)}\,\frac{r^{(k)}_{1,-}}{r^{(k)}_{1,-}+j}\,e^{-jh_0}\\ \displaystyle\sum_{j=0}^{k}\frac{c_{k,j}(h_0)}{q-\psi^{(k)}_Z(j)}\,\frac{r^{(k)}_{2,-}}{r^{(k)}_{2,-}+j}\,e^{-jh_0}\end{pmatrix}.$$

*By Lemma 5.4 below, $\mathbf{M}^{(k)}_{\mathrm{bar}}(q)$ is invertible, giving*

$$\mathbf{B}^{(k)}(q;h_0)=-\big(\mathbf{M}^{(k)}_{\mathrm{bar}}(q)\big)^{-1}\mathbf{b}^{(k)}(q;h_0).$$

*Proof. Step 1: boundary conditions from killing.* The killing condition $\widehat{U}(q,z;h_0)=0$ for $z\le -h_0$ imposes, at $z=-h_0^+$,

(i) $\widehat{U}(q,-h_0^+;h_0)=0$;

(ii) $K^{(k)}_{j,-}\left[\widehat{U}(q,\cdot;h_0)\right](-h_0^+)=0,\quad j=1,2.$

Condition (i) is continuity at the barrier. Conditions (ii) arise because $K^{(k)}_{j,-}[f](z)=\int_{-\infty}^{0}f(z+y)\,r^{(k)}_{j,-}e^{r^{(k)}_{j,-}y}\mathrm{d}y$ integrates $f$ over landing points $z+y<z$; for $z=-h_0^+$ every landing point satisfies $z+y<-h_0$, where $\widehat{U}=0$, so the integral vanishes.

*Step 2: substitute the full solution.* Write $\widehat{U}=\widehat{U}_{\mathrm{part}}+\widehat{U}_{\mathrm{hom}}$ and insert into each condition. Since the conditions are linear, each splits as

$$[\text{condition on }\widehat{U}_{\mathrm{hom}}]=-[\text{condition on }\widehat{U}_{\mathrm{part}}].$$

*Step 3: condition (i).* Evaluating at $z=-h_0$ and using $e^{\gamma_{m+3}(q)\cdot 0}=1$,

$$\widehat{U}_{\mathrm{hom}}(q,-h_0;h_0)=\sum_{m=1}^{3}B^{(k)}_m=-\widehat{U}_{\mathrm{part}}(q,-h_0;h_0),$$

which is the first row of the system.

*Step 4: conditions (ii).* Apply $K^{(k)}_{j,-}$ to $\widehat{U}_{\mathrm{hom}}$. By the phase-operator formula (17),

$$K^{(k)}_{j,-}\left[e^{\gamma_{m+3}(\cdot+h_0)}\right](-h_0)=\frac{r^{(k)}_{j,-}}{r^{(k)}_{j,-}+\gamma_{m+3}(q)}\,e^{\gamma_{m+3}(q)\cdot 0}=\frac{r^{(k)}_{j,-}}{r^{(k)}_{j,-}+\gamma_{m+3}(q)},$$

giving row $j+1$ of the system:

$$\sum_{m=1}^{3} \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma_{m+3}(q)} B_m^{(k)} = -K_{j,-}^{(k)} \left[\widehat{U}_{\text{part}}(q,\cdot;h_0)\right](-h_0).$$

Applying (17) to $\widehat{U}_{\text{part}}(q,z;h_0) = \sum_j \frac{c_{k,j}}{q-\psi_Z^{(k)}(j)} e^{jz}$ yields the explicit entries of $\mathbf{b}^{(k)}$ stated above.

*Step 5: matrix form and solution.* The three conditions assemble into $\mathbf{M}_{\text{bar}}^{(k)}(q)\,\mathbf{B}^{(k)} = -\mathbf{b}^{(k)}$. By Lemma 5.4, $\mathbf{M}_{\text{bar}}^{(k)}(q)$ is invertible, so

$$\mathbf{B}^{(k)}(q;h_0) = -\left(\mathbf{M}_{\text{bar}}^{(k)}(q)\right)^{-1} \mathbf{b}^{(k)}(q;h_0). \qquad \square$$

**Lemma 5.4** (Invertibility of $\mathbf{M}_{\text{bar}}^{(k)}(q)$)**.** *Under Assumption 5.1 and condition* (24)*, the barrier matrix* $\mathbf{M}_{\text{bar}}^{(k)}(q)$ *of Proposition 5.3 is invertible.*

*Proof.* Subtracting row 1 from rows 2 and 3 replaces entry $(i,m)$ for $i=2,3$ by $-\gamma_{m+3}(q)/(r_{i-1,-}^{(k)} + \gamma_{m+3}(q))$. Pulling out $-1$ from each of these two rows (net sign $(-1)^2 = 1$) and then factoring $\gamma_{m+3}(q)$ from each column gives

$$\det\bigl(\mathbf{M}_{\text{bar}}^{(k)}\bigr) = \gamma_4\gamma_5\gamma_6 \, \det\left(\frac{1}{x_i + \gamma_{j+3}(q)}\right)_{1\le i,j\le 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i + \gamma_{j+3}}\right) = \frac{\prod_{1\le i<i'\le 3}(x_{i'} - x_i)\cdot \prod_{4\le m<m'\le 6}(\gamma_{m'} - \gamma_m)}{\prod_{i=1}^{3}\prod_{m=4}^{6}(x_i + \gamma_m)}.$$

Since $x_1 = 0$, the factor $\prod_{m=4}^{6}(x_1+\gamma_m) = \gamma_4\gamma_5\gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (24); the $\gamma_m$ are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. $\square$

Evaluating $\widehat{U}$ at the initial state $Z_0 = 0$ gives

$$\widehat{U}(q,0;h_0) = \sum_{j=0}^{k} \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} + \sum_{m=1}^{3} B_m^{(k)}(q;h_0)\, e^{\gamma_{m+3}(q)h_0}. \tag{25}$$

The killed moment is then recovered by Laplace inversion:

$$m_k(T;h_0) = \mathscr{L}^{-1}_{q\to T}\left[\widehat{U}(q,0;h_0)\right].$$

### *5.2. Survival Probability*

The same resolvent architecture yields the survival probability. Define

$$p_{\text{surv}}(t,z;h_0) := \mathbb{P}_z(\tau(h_0) > t), \qquad p_{\text{liq}}(t,z;h_0) := 1 - p_{\text{surv}}(t,z;h_0).$$

Let $\mathscr{L}^{(0)}$ denote the generator of $Z$ under the untilted measure, with exponent $\psi_Z^{(0)}$, and define

$$\widehat{U}_{\text{surv}}(q,z;h_0) := \int_0^\infty e^{-qt} p_{\text{surv}}(t,z;h_0)\,\mathrm{d}t.$$

The problem is

$$(q-\mathscr{L}^{(0)})\widehat{U}_{\text{surv}}(q,z;h_0)=1, \qquad z>-h_0,$$

with

$$\widehat{U}_{\text{surv}}(q,z;h_0)=0, \qquad z\le -h_0.$$

The domain is the half-line $(-h_0,\infty)$, the boundary is absorbing at $z=-h_0$, and there is no interface because the source is constant.

**Proposition 5.5** (Survival resolvent and liquidation probability)**.** *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(0)}\neq r_{2,-}^{(0)}, \quad \textit{i.e.,} \quad \eta_{1,-}\neq\eta_{2,+}, \tag{26}$$

*let $\gamma_4^{(0)}(q),\gamma_5^{(0)}(q),\gamma_6^{(0)}(q)$ be the three roots of $\psi_Z^{(0)}(\gamma)=q$ with negative real part. Then*

$$\widehat{U}_{\text{surv}}(q,z;h_0)=\frac{1}{q}+C_1(q;h_0)e^{\gamma_4^{(0)}(q)(z+h_0)}+C_2(q;h_0)e^{\gamma_5^{(0)}(q)(z+h_0)}+C_3(q;h_0)e^{\gamma_6^{(0)}(q)(z+h_0)},$$

*where the coefficients $\mathbf{C}(q;h_0):=(C_1,C_2,C_3)^\top$ satisfy the linear system*

$$\mathbf{M}_{\text{surv}}(q)\,\mathbf{C}(q;h_0)=-\begin{pmatrix}1/q\\1/q\\1/q\end{pmatrix},$$

*with the $3\times3$ survival matrix*

$$\mathbf{M}_{\text{surv}}(q)=\begin{pmatrix}1 & 1 & 1\\ \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)}+\gamma_4^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)}+\gamma_5^{(0)}(q)} & \dfrac{r_{1,-}^{(0)}}{r_{1,-}^{(0)}+\gamma_6^{(0)}(q)}\\ \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)}+\gamma_4^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)}+\gamma_5^{(0)}(q)} & \dfrac{r_{2,-}^{(0)}}{r_{2,-}^{(0)}+\gamma_6^{(0)}(q)}\end{pmatrix}.$$

*By Lemma 5.6 below, $\mathbf{M}_{\text{surv}}(q)$ is invertible, giving*

$$\mathbf{C}(q;h_0)=-\big(\mathbf{M}_{\text{surv}}(q)\big)^{-1}\begin{pmatrix}1/q\\1/q\\1/q\end{pmatrix}.$$

*At the initial state $Z_0=0$,*

$$\widehat{U}_{\text{surv}}(q,0;h_0)=\frac{1}{q}+C_1(q;h_0)e^{\gamma_4^{(0)}(q)h_0}+C_2(q;h_0)e^{\gamma_5^{(0)}(q)h_0}+C_3(q;h_0)e^{\gamma_6^{(0)}(q)h_0}.$$

*Hence*

$$p_{\text{surv}}(T,0;h_0)=\mathscr{L}^{-1}_{q\to T}\Big[\widehat{U}_{\text{surv}}(q,0;h_0)\Big], \qquad p_{\text{liq}}(T,0;h_0)=1-p_{\text{surv}}(T,0;h_0).$$

*Proof. Step 1: resolvent equation.* Taking the Laplace transform of the PIDE $\partial_t p_{\text{surv}}=\mathscr{L}^{(0)}p_{\text{surv}}$ with initial condition $p_{\text{surv}}(0,z;h_0)=1$ and killing condition $p_{\text{surv}}=0$ for $z\le -h_0$, exactly as in Section 5 with $g_k\equiv 1$, gives

$$(q-\mathscr{L}^{(0)})\widehat{U}_{\text{surv}}(q,z;h_0)=1, \qquad z>-h_0.$$

*Step 2: particular solution.* The constant forcing $f \equiv 1$ is an eigenfunction of $\mathscr{L}^{(0)}$ with eigenvalue $\psi_Z^{(0)}(0) = 0$, so the particular solution is

$$\widehat{U}_{\text{surv,part}}(q,z) = \frac{1}{q - \psi_Z^{(0)}(0)} = \frac{1}{q}.$$

*Step 3: homogeneous solution.* By Lemma 5.2 applied with $k = 0$, the admissible homogeneous solution on $z > -h_0$ is

$$\widehat{U}_{\text{surv,hom}}(q,z;h_0) = C_1 e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2 e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3 e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where $\gamma_4^{(0)}, \gamma_5^{(0)}, \gamma_6^{(0)}$ are the roots of $\psi_Z^{(0)}(\gamma) = q$ with $\operatorname{Re}(\gamma) < 0$.

*Step 4: boundary conditions.* The killing condition imposes, at $z = -h_0^+$:

(i) $\widehat{U}_{\text{surv}}(q, -h_0^+; h_0) = 0$;

(ii) $K_{j,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0, \quad j = 1, 2,$

by the same argument as in Proposition 5.3: downward jumps from $-h_0^+$ land in the killed region.

*Step 5: linear system.* Substituting the full solution $\widehat{U}_{\text{surv}} = 1/q + \widehat{U}_{\text{surv,hom}}$ into each condition and applying (17) to the homogeneous terms yields

$$\mathbf{M}_{\text{surv}}(q)\,\mathbf{C}(q;h_0) = -\begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

where the right-hand side entries are $K_{j,-}^{(0)}[1/q](-h_0^+) = 1/q$ (since $K_{j,-}^{(0)}$ integrates to one against the exponential density). By Lemma 5.6 below, $\mathbf{M}_{\text{surv}}(q)$ is invertible, so the coefficients $\mathbf{C}$ are uniquely determined.

*Step 6: inversion and liquidation probability.* Evaluating at $Z_0 = 0$ and inverting via the Bromwich formula gives $p_{\text{surv}}(T, 0; h_0)$. The liquidation probability follows from the complement. □

**Lemma 5.6** (Invertibility of $\mathbf{M}_{\text{surv}}(q)$)**.** *Under Assumption 5.1 and condition* (26)*, the survival matrix* $\mathbf{M}_{\text{surv}}(q)$ *of Proposition 5.5 is invertible.*

*Proof.* The argument is identical to that of Lemma 5.4 with $k = 0$. The phase rates are $r_{1,-}^{(0)} = 1/\eta_{1,-}$ and $r_{2,-}^{(0)} = 1/\eta_{2,+}$, so condition (26) is equivalent to $\eta_{1,-} \neq \eta_{2,+}$. Each denominator factor is nonzero because $-r_{j,-}^{(0)}$ is a pole of $\psi_Z^{(0)}$, hence cannot equal any root $\gamma_m^{(0)}(q)$ of $\psi_Z^{(0)}(\cdot) = q$ for finite $q$. □

## 6. Numerical Results

We validate the Laplace–resolvent method against Monte Carlo (MC) simulation and verify the analytical no-shorting limit as $h_0 \to \infty$. Time-domain quantities are recovered via the Talbot deformed-contour algorithm (Talbot, 1979). No Bromwich shift needs to be chosen by hand.

The Kou model and contract parameters are given in Table 3. The two assets are assigned distinct downward-jump means ($\eta_{1,-} = 0.08$, $\eta_{2,-} = 0.12$), which ensures the phase rates $r_{1,-}^{(0)} \neq r_{2,-}^{(0)}$ and the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}$ is non-singular for all admissible $k$ (here $k_{\max} = \lfloor 1/0.09 \rfloor - 1 = 10$).

| Group | Parameter | Asset 1 | Asset 2 |
|---|---|---|---|
| Kou | Expected price growth $\mu_i$ | 0.10 | 0.08 |
| | Diffusion volatility $\sigma_i$ | 0.30 | 0.25 |
| | Jump intensity $\lambda_i$ | 2.0 | 2.0 |
| | Upward-jump probability $p_i$ | 0.50 | 0.50 |
| | Mean upward-jump size $\eta_{i,+}$ | 0.10 | 0.09 |
| | Mean downward-jump size $\eta_{i,-}$ | 0.08 | 0.12 |
| Correlation / contract | Brownian correlation $\rho$ | 0.50 | |
| | Collateral factor $b$ | 0.80 | |
| Portfolio | Initial prices $S_{1,0} = S_{2,0}$ | 1.00 | |
| | Horizon $T$ | 1.00 | |

Table 3: Kou model and contract parameters. Derived quantities: $\chi_1 \approx 0.0185$, $\chi_2 \approx 0.0172$, $\mu_1^{\mathrm{eff}} \approx 0.0180$, $\mu_2^{\mathrm{eff}} \approx 0.0092$.

For each $h_0 \in \{0.01, 0.10, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00\}$ we report the survival probability $p_{\mathrm{surv}}$ and the following conditional statistics of $\Pi_T \mid \{\tau > T\}$, where $\mu_k := \mathbb{E}[\Pi_T^k \mid \tau > T]$:

$$
\begin{aligned}
&\text{mean} := \mu_1, \qquad \text{variance} := \mu_2 - \mu_1^2, \\
&\text{skewness} := \frac{\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3}{(\mu_2 - \mu_1^2)^{3/2}}, \qquad \text{ex. kurt} := \frac{\mu_4 - 4\mu_1\mu_3 + 6\mu_1^2\mu_2 - 3\mu_1^4}{(\mu_2 - \mu_1^2)^2} - 3.
\end{aligned}
\tag{27}
$$

As $h_0 \to \infty$ the short holding $n_2 \to 0$ and the portfolio converges to a unit long position in asset 1 (Remark 3), so all conditional statistics converge to those of $e^{X_{1,T}}$ under $\mathbb{P}$:

$$
\lim_{h_0 \to \infty} \mu_k = \mathbb{E}_{\mathbb{P}}\left[e^{kX_{1,T}}\right] = \exp\bigl(T\,\Psi(-ik, 0)\bigr).
$$

Table 4 reports all statistics together with these analytical limits.

### *6.1. Parameter sensitivity*

Figure 1 shows how $p_{\mathrm{surv}}$ (left column) and the conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$ (right column) vary with $h_0$ as each parameter is swept in turn. For the collateral factor $b$ (row e–f), survival probability is independent of $b$ at fixed $h_0$ — only the payoff kernel changes — so the left panel shows the initial leverage $L_0 = e^{h_0}/(e^{h_0} - b)$ instead.

Three patterns dominate. (i) *Volatility* is the strongest driver: at $0.3\times$ scale, almost all paths survive for $h_0 \geq 0.3$; at $2\times$ scale, the same cushion yields survival below 40%. (ii) *Correlation* acts entirely through the effective variance of $Z = X_1 - X_2$: high $\rho$ narrows the spread of $Z$ and lifts the survival curve, with negligible effect on the conditional mean. (iii) *Jump intensity* and *horizon* compress survival monotonically, while increasing the conditional mean at low $h_0$ through selective survivor effects. For $b$, leverage can exceed $10\times$ at $h_0 < 0.2$ for $b = 0.95$, explaining why the conditional mean is highly sensitive to $b$ at small cushions.

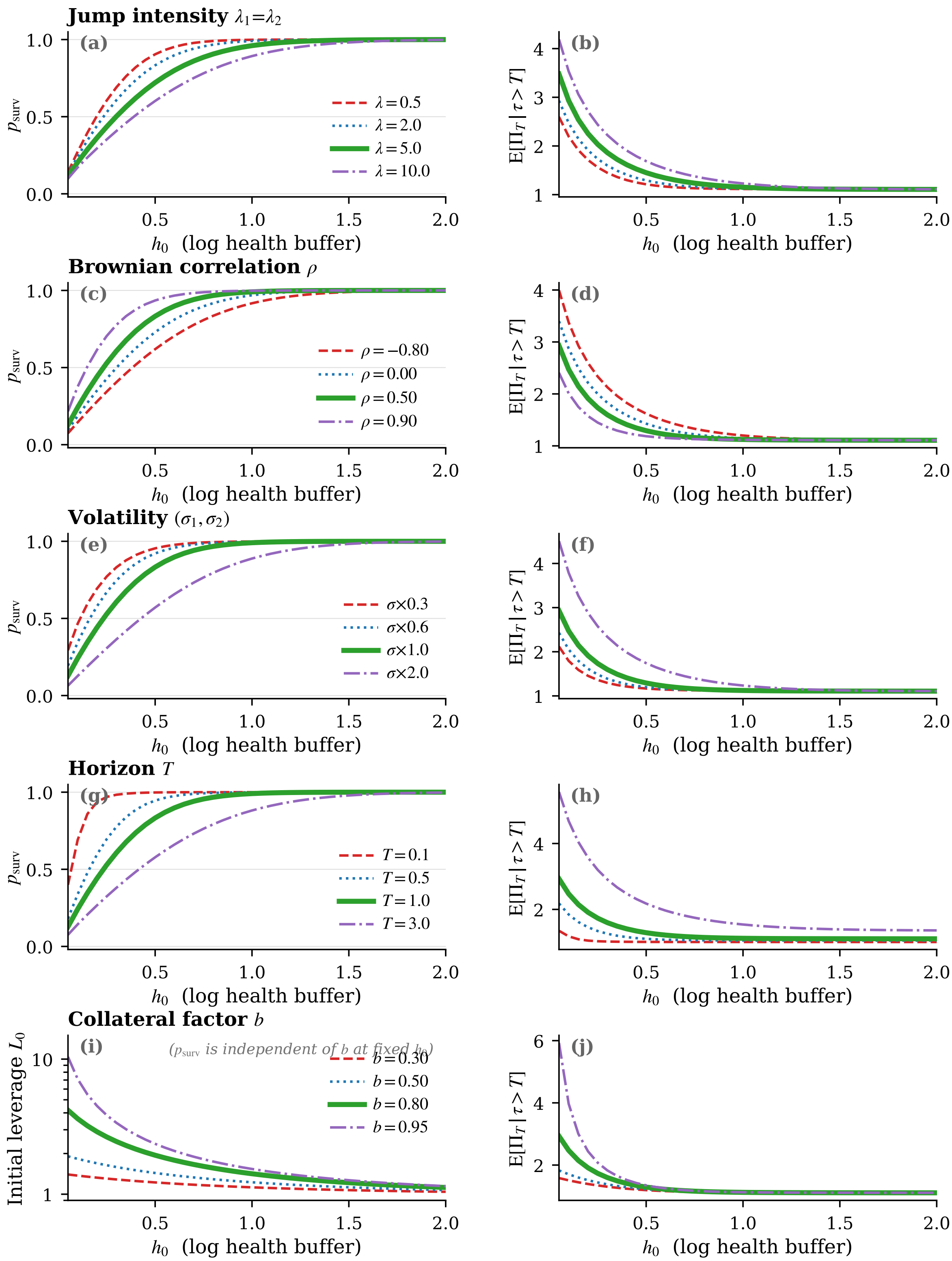


Figure 1: Left column: survival probability $p_{\text{surv}}$ (rows a–d) and initial leverage $L_0$ (row e, log scale) versus $h_0$. Right column: conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$. Each row varies one parameter; all others fixed at Table 3. Encoding: colour and line style identify parameter values (solid/green = benchmark); the heavier line is the baseline. Laplace–resolvent, $h_0 \in [0.05, 2.00]$, 40 points per curve.

| $h_0$ | $p_{\text{surv}}$ Lap. | $p_{\text{surv}}$ MC | Mean | Variance | Skewness | Ex. Kurt. |
|---|---|---|---|---|---|---|
| 0.01 | 0.0271 | 0.0550 | 3.4548 | 3.5915 | 2.0297 | 9.3254 |
| 0.10 | 0.2420 | 0.2631 | 2.4707 | 1.8218 | 2.0057 | 9.0864 |
| 0.25 | 0.5290 | 0.5444 | 1.7329 | 0.9009 | 1.9462 | 8.5055 |
| 0.50 | 0.8331 | 0.8412 | 1.2922 | 0.4955 | 1.7843 | 7.1965 |
| 0.75 | 0.9560 | 0.9586 | 1.1602 | 0.3648 | 1.6254 | 6.1832 |
| 1.00 | 0.9908 | 0.9917 | 1.1231 | 0.2975 | 1.5394 | 5.7212 |
| 1.50 | 0.9998 | 0.9998 | 1.1102 | 0.2289 | 1.5019 | 5.4775 |
| 2.00 | 1.0000 | 1.0000 | 1.1078 | 0.1992 | 1.4897 | 5.3480 |
| $\infty$ | 1.0000 | — | 1.1052 | 0.1668 | 1.4493 | 5.0371 |

Table 4: Laplace–resolvent conditional statistics and the no-shorting analytical limit. Mean, variance, skewness, and excess kurtosis are conditional on survival ($\tau > T$) and computed via the resolvent method (Laplace); survival probability MC benchmark uses $10^5$ Euler–Poisson paths (252 steps, seed 42). The large MC discrepancy in $p_{\text{surv}}$ at $h_0 = 0.01$ reflects path scarcity ($\approx$2,700 surviving paths), where MC is unreliable; the Laplace method is unaffected. The bottom row gives the analytical limits as $h_0 \to \infty$; Laplace at $h_0 = 100$ matches these values to relative error $< 10^{-11}$, confirming internal consistency of the resolvent construction, Talbot inversion, and the drift normalisation $\mu_i^{\text{eff}} = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i \chi_i$.

## 7. Discussion

### *7.1. Model limitations*

The model yields an exact solution within a deliberately narrow scope. The following maintained assumptions represent the primary directions for future work.

**Instantaneous oracle and price discovery.** The health factor $H_t$ in (1) is computed from protocol price oracles that report asset prices with a lag — typically one block (12 seconds on Ethereum mainnet). During periods of extreme volatility the oracle price can deviate substantially from the true market price, allowing the true health factor to breach one before the protocol detects it, or preventing liquidation when the true health factor has in fact recovered. The model treats oracle prices as continuous and perfectly synchronised with the underlying asset processes.

**Zero slippage at liquidation.** The absorbing-barrier assumption implies that collateral is liquidated at the mid-price prevailing at $\tau$. In practice a keeper must sell the collateral into on-chain liquidity pools, and the effective execution price is depressed by slippage on that trade. For large positions this effect can be material; Qin et al. (2021) document cases where slippage exceeded the liquidation bonus itself.

**No liquidation bonus recovery.** The zero-recovery proxy (Remark 1) ignores the keeper's liquidation bonus, which reduces the borrower's net recovery. Incorporating an explicit recovery fraction $1-\ell$ on liquidated paths — where $\ell$ is the protocol bonus rate — would add a second term to $\Phi$ in (5) but would not alter the structure of the resolvent, since the bonus affects only paths on $\{\tau \leq T\}$.

**Static collateral management.** The investor posts an initial collateral position and makes no further adjustments before $T$. A sophisticated participant may top up collateral or partially unwind the short leg as $H_t$ approaches one, dynamically managing the distance to the boundary. Incorporating such controls would convert the problem into a stochastic optimal control problem over a reflected or absorbed process.

**Deterministic interest rates.** DeFi lending protocols charge a floating borrow rate updated algorith-

|  | Laplace − MC |  |  |  |  | CPU time (s) |  |
|---|---|---|---|---|---|---|---|
| $h_0$ | $\Delta p_{\text{surv}}$ | $\Delta$Mean | $\Delta$Var. | $\Delta$Skew. | $\Delta$Kurt. | Lap. | MC |
| 0.01 | −0.0278 | 0.0186 | −0.1502 | −0.0709 | 1.3105 | 0.06 | 3.38 |
| 0.10 | −0.0210 | 0.0271 | −0.0120 | −0.0307 | 0.8078 | 0.06 | 3.29 |
| 0.25 | −0.0154 | 0.0130 | −0.0043 | −0.0155 | 0.6354 | 0.06 | 3.32 |
| 0.50 | −0.0081 | 0.0060 | −0.0041 | −0.0182 | 0.2256 | 0.06 | 3.29 |
| 0.75 | −0.0026 | 0.0012 | −0.0023 | −0.0195 | 0.1239 | 0.06 | 3.30 |
| 1.00 | −0.0008 | −0.0004 | −0.0015 | −0.0227 | 0.0731 | 0.06 | 3.31 |
| 1.50 | −0.0001 | −0.0011 | −0.0010 | −0.0256 | 0.0378 | 0.06 | 3.31 |
| 2.00 | 0.0000 | −0.0012 | −0.0009 | −0.0268 | 0.0218 | 0.06 | 3.32 |

Table 5: Laplace minus MC differences ($\Delta = \text{Lap.} - \text{MC}$) and wall-clock computation time. Differences shrink as $h_0$ increases because Monte Carlo becomes more reliable when many paths survive. The large $\Delta$Kurt at $h_0 = 0.01$ reflects MC noise from path scarcity ($\approx$2,700 surviving paths out of $10^5$); $\Delta p_{\text{surv}}$ is negative throughout because MC over-counts surviving paths at finite sample sizes. The Laplace method covers five Talbot inversions ($k = 0, \ldots, 4$); the $\approx 55\times$ speed-up over MC is uniform across all $h_0$ values.

mically as a function of pool utilisation (Gudgeon et al., 2020). The model absorbs the funding cost into the effective drift $\mu_2^{\text{eff}}$, treating the rate as deterministic and fixed at inception. Stochastic interest rates would couple the short-leg dynamics to a latent utilisation process.

**No funding constraints.** The analysis assumes the investor can hold the initial position to horizon $T$ without receiving collateral top-up demands. A capital-constrained investor who cannot post additional margin when required may be forced to close the position before $\tau$, introducing an additional stopping time that is not captured by the model.

### *7.2. Practical implementation*

Despite the above limitations, the semi-analytic framework provides a tractable and fast starting point for position sizing. A natural two-step procedure is the following.

**Step 1 — survival constraint.** Fix a maximum tolerated liquidation probability $\bar{p}$ over the horizon $T$, for example $\bar{p} = 5\%$. The survival probability $p_{\text{surv}}(h_0)$ is monotonically increasing in $h_0$ (Table 4): at $h_0 = 0.50$ survival already exceeds 83% under the benchmark parameters. Inverting this map identifies the minimum admissible health buffer $\underline{h}_0$ satisfying $p_{\text{liq}}(T, 0; \underline{h}_0) \leq \bar{p}$. The Laplace method evaluates $p_{\text{surv}}$ in $\approx 60\,\text{ms}$, so the inversion is a fast scalar root-find.

**Step 2 — conditional upside maximisation.** Over the feasible set $h_0 \geq \underline{h}_0$, maximise the conditional expected payoff, or a mean–variance functional, using the semi-analytic moments. Because the moments are noise-free and differentiable in $h_0$, a standard one-dimensional line search converges in a handful of evaluations with no simulation variance.

The output of this procedure is the *optimal initial health buffer* $h_0^\star$: the smallest cushion above the liquidation boundary that simultaneously satisfies the survival constraint and maximises conditional performance. The corresponding initial leverage is then read off from (4) as $L_0^\star = e^{h_0^\star}/(e^{h_0^\star} - b)$; leverage is a consequence of the health-buffer choice, not the primary control variable.

Figure 2 visualises the full tradeoff in three dimensions. The vertical axis is the conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$; the two horizontal axes are the conditional variance and survival probability. Each curve is parametrised by $h_0 \in [0.04, 2.00]$: as $h_0$ falls, conditional variance rises and the curve moves forward along the $p_{\text{surv}}$ axis toward lower survival, while the conditional mean (height) first rises then flattens. The floor projection (dotted) anchors the curve to the variance–survival plane; vertical drop lines link four labelled $h_0$ values to their floor shadows. The two flanking frontiers show how scaling volatility by $0.5\times$ or $1.5\times$ shifts the entire surface: lower volatility collapses the curve toward low variance and high survival; higher volatility expands it, raising conditional upside at the cost of heavier liquidation risk.

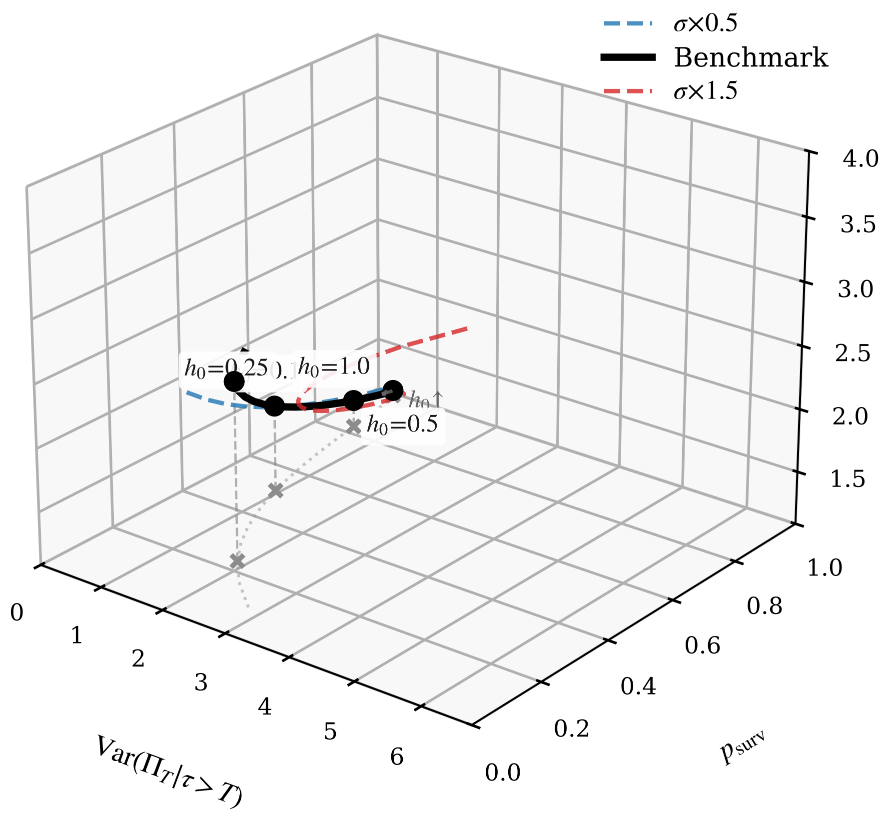


Figure 2: Mean–variance–survival frontier of $\Pi_T \mid \{\tau > T\}$ for three volatility regimes ($\sigma \times 0.5$ blue dashed, benchmark black solid, $\sigma \times 1.5$ red dashed). Axes: $x = \mathrm{Var}(\Pi_T \mid \tau > T)$, $y = p_{\text{surv}}$, $z = \mathrm{E}[\Pi_T \mid \tau > T]$ (vertical). Each curve is parametrised by $h_0 \in [0.04, 2.00]$; the arrow marks the direction of increasing $h_0$ (lower leverage, lower risk, lower conditional upside). Four benchmark points are annotated with their $h_0$; vertical drop lines anchor them to the floor projection (dotted grey), which shows the variance–survival plane at the minimum conditional mean. Script `jobs/frontier_analysis.py`.

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## Appendix A: Generalised Beta of the Second Kind Approximation to the Payoff Distribution

The semi-analytic machinery of Section 5 delivers the first four killed moments $M_k(h_0)$, $k = 1,2,3,4$, and the survival probability $p_{\text{surv}}(h_0)$. This appendix shows how to fit a four-parameter *Generalised Beta of the Second Kind* (GB2) distribution to these outputs.

On $\{\tau > T\}$ the payoff satisfies

$$Y := \Pi_T \mid \{\tau > T\} = \frac{n_2}{b} S_{2,0}\, e^{X_{2,T}} \,(e^{h_0+Z_T} - b).$$

Since $Z_T > -h_0$ on $\{\tau > T\}$, we have $e^{h_0+Z_T} > 1 > b$, so the factor $(e^{h_0+Z_T} - b) > 0$. However, because $e^{X_{2,T}}$ can be arbitrarily close to zero, $\inf Y = 0$, and the support is $Y \in (0,\infty)$ with no natural positive shift. The full payoff law is

$$\Pi_T \overset{d}{=} p_{\text{liq}}\, \delta_0 + p_{\text{surv}}\, Y, \tag{28}$$

where $p_{\text{liq}} = 1 - p_{\text{surv}}$.

Since $\Pi_T^k = Y^k\, \mathbf{1}_{\{\tau>T\}}$,

$$\mu_k^+ := \mathbb{E}[Y^k] = \frac{M_k(h_0)}{p_{\text{surv}}(h_0)}, \qquad M_k(h_0) = \frac{e^{T\Psi(0,-ik)}}{(e^{h_0} - b)^k}\, m_k(T;h_0), \tag{29}$$

from Proposition 4.1, for $k = 1,2,3,4$.

We model $Y \sim \text{GB2}(a,\beta,p,q)$ with density

$$f_Y(y) = \frac{a\, y^{ap-1}}{\beta^{ap} B(p,q) \left(1 + (y/\beta)^a\right)^{p+q}}, \qquad y > 0, \tag{30}$$

where $a > 0$ is a tail/shape parameter, $\beta > 0$ is a scale parameter, $p > 0$ and $q > 0$ are left- and right-shape parameters, and $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ is the beta function.

**Lemma A.1** (GB2 raw moments)**.** *For $k < aq$, the k-th raw moment of $Y \sim \mathrm{GB2}(a,\beta,p,q)$ is*

$$\mathbb{E}[Y^k] = \beta^k \frac{B(p+k/a,\ q-k/a)}{B(p,q)} = \beta^k \frac{\Gamma(p+k/a)\,\Gamma(q-k/a)}{\Gamma(p)\,\Gamma(q)}. \tag{31}$$

*The first four moments exist if and only if $q > 4/a$.*

We fit $(a,\beta,p,q)$ by matching the four conditional moments $\mu_1^+,\dots,\mu_4^+$.

*Step 1: shape parameters from scale-free ratios.* Define

$$R_k := \frac{\mu_k^+}{(\mu_1^+)^k} = \frac{B(p+k/a,\ q-k/a)/B(p,q)}{\big[B(p+1/a,\ q-1/a)/B(p,q)\big]^k}, \qquad k=2,3,4. \tag{32}$$

The ratios $R_k$ depend only on $(a,p,q)$, not on $\beta$. Solve

$$R_k(a,p,q) = \hat{R}_k, \qquad k=2,3,4, \tag{33}$$

where $\hat{R}_k = \mu_k^+/(\mu_1^+)^k$ are the targets, subject to $a>0$, $p>0$, $q>4/a$. For numerical stability, evaluate (32) in log-space:

$$\log \mathbb{E}[Y^k] = k\log\beta + \log\Gamma\big(p+\tfrac{k}{a}\big) + \log\Gamma\big(q-\tfrac{k}{a}\big) - \log\Gamma(p) - \log\Gamma(q). \tag{34}$$

*Step 2: scale from the first moment.* Given $(a,p,q)$ from Step 1, recover

$$\beta = \mu_1^+ \frac{B(p,q)}{B(p+1/a,\ q-1/a)}. \tag{35}$$

The CDF of $\Pi_T$ is

$$F_\Pi(y) = \begin{cases} 0 & y<0, \\ p_{\text{liq}} & y=0, \\ p_{\text{liq}} + p_{\text{surv}}\,F_Y(y) & y>0, \end{cases}$$

where $F_Y$ is the GB2 CDF. For a confidence level $u \in (0,1)$, the Value-at-Risk is

$$\mathrm{VaR}(u) = \begin{cases} 0 & u \le p_{\text{liq}}, \\ F_Y^{-1}\!\left(\dfrac{u-p_{\text{liq}}}{p_{\text{surv}}}\right) & u > p_{\text{liq}}, \end{cases}$$

where $F_Y^{-1}$ is the GB2 quantile function.

After fitting, verify that

$$\beta^k \frac{\Gamma(p+k/a)\,\Gamma(q-k/a)}{\Gamma(p)\,\Gamma(q)} = \mu_k^+, \qquad k=1,2,3,4.$$

The GB2 family nests several common distributions (Burr XII, log-logistic, Pareto, Weibull, log-normal in a limit) and has four free parameters, enabling the simultaneous matching of four conditional moments. It accommodates the power-law tails that arise in Kou jump-diffusion models, which the Generalized Gamma family — having thinner stretched-exponential tails — cannot capture. The condition $q > 4/a$ provides an explicit testable constraint on whether the fourth moment of the conditional payoff is finite.